

Where the CCX branch phases come from, and how to handle them

Diagnosis of the V_{cc} -induced phase mismatch in `ccx_pedersen_average_fidelity.py`, with comparison to Farouk et al. (Photonics 2023). Updated for the current `pulses.py` convention.

TriQG · examples/Average_fidelity_Vcc_newenergy · 2026-04-27

TL;DR. Both `ccx_pedersen_average_fidelity.py` and `or_pedersen_average_fidelity.py` now run with the **physical** $V_{cc}/(2\pi) = -5.54$ MHz at $a = 5$ μm (Option 0 also active: `triqq/pulses.py:omega_t2` returns $-\text{amp}/2$). Verified results:

Metric	CCX (40 ns)	OR (320 ns)	What it is
Survival T_P	0.996744	0.996173	population stayed in computational subspace
Basis fidelity $\overline{F}_{\text{basis}}$	0.996620	0.992611	Yu et al. Eq. (7) – mean per-input fidelity
Pedersen raw $\overline{F}$	0.796283	0.180560	Pedersen Eq. (5) against the bare U_0
Pedersen phase-corrected $\overline{F}$	0.996997	0.992572	Pedersen Eq. (5) against DU_0 , D diagonal

For the paper, report the bold numbers (phase-corrected Pedersen average gate fidelity), with explicit specification of the diagonal correction D . This is directly comparable to Farouk et al.'s 99.7% for the EIT $C_2\text{NOT}_2$ gate. The full publication-ready text and formula derivation are in Section 1.

Why $\overline{F}_{\text{raw}}$ looks bad: the $V_{cc} \approx -5.54$ MHz Cs–Cs interaction adds a known coherent CZ phase between the two control atoms during the gate. The bare Pedersen formula penalises it; the phase-corrected variant absorbs it as a virtual two-qubit Z that the surrounding circuit handles for free. See Section 2 for what each metric measures.

Both gates implement $\mathcal{E} \approx D \cdot U_0$ where U_0 is the ideal Toffoli (CCX) or OR gate and D is a diagonal unitary on the 8-dim computational basis – mostly virtual-Z's plus one $\text{CZ}_{c_1 c_2}$ from V_{cc} . The unfixable infidelity (decoherence + leakage) is $1 - \overline{F}_{\text{PC}} \approx 3 \times 10^{-3}$ for CCX and $\approx 7.4 \times 10^{-3}$ for OR.

1 Paper-ready Pedersen fidelity report

This section is designed to be cut-and-pasted into your manuscript with minimal editing. It contains the formula derivation, both gate's numerical results at the physical $V_{cc} = -5.54$ MHz, and draft Methods + Results paragraphs.

1.1 The Pedersen average gate fidelity

Definition. For a CPTP map $\mathcal{E}$ acting on a d -dimensional quantum system and a target unitary U_0 , the average gate fidelity is the mean output-state overlap with the ideal output, averaged over Haar-random pure input states:

$$\overline{F}(\mathcal{E}, U_0) = \int d\psi \langle \psi | U_0^\dagger \mathcal{E}(|\psi\rangle\langle\psi|) U_0 | \psi \rangle. \quad (1)$$

Closed form (Pedersen et al. 2007). This Haar integral evaluates in closed form to

$$\overline{F}(\mathcal{E}, U_0) = \frac{d F_{\text{pro}}(\mathcal{E}, U_0) + 1}{d + 1} \quad (\text{Pedersen Eq. (5)}) \quad (2)$$

where F_{pro} is the **process (entanglement) fidelity**,

$$F_{\text{pro}}(\mathcal{E}, U_0) = \frac{1}{d^2} \sum_{i,j=1}^d \langle \pi(i), \mathcal{E}(\rho_{ij}), \pi(j) \rangle \quad (\text{Pedersen Eq. (3)}) \quad (3)$$

Here $\{|\psi_i\rangle\}_{i=1}^d$ is any orthonormal basis of the computational subspace and $|\pi(i)\rangle = U_0 |\psi_i\rangle$ the ideal output. For a noiseless unitary channel $\mathcal{E}(\rho) = U_{\text{eff}} \rho U_{\text{eff}}^\dagger$, Equation 3 reduces to

$$F_{\text{pro}}^{\text{unitary}}(U_{\text{eff}}, U_0) = 1/d^2 \left| \text{Tr}(U_0^\dagger U_{\text{eff}}) \right|^2. \quad (4)$$

1.2 Computing it for the TriQG simulation

The simulator's full Hilbert space is 36-dimensional ($\text{Cs} \times \text{Cs} \times \text{Rb} = 3 \times 3 \times 4$), but the gate acts on the $d = 2^3 = 8$ -dimensional computational subspace $\mathcal{S} = \text{span}\{|c_1, c_2, t\rangle : c_i \in \{0, 1\}, t \in \{A, B\}\}$. Let V be the 36×8 isometry whose columns are the eight computational kets in the full simulator basis. The Choi tensor restricted to $\mathcal{S}$ is

$$C[i, j, k, l] = \langle \psi_k | \mathcal{E}(|\psi_i\rangle\langle\psi_j|) | \psi_l \rangle, \quad i, j, k, l = 1, \dots, 8, \quad (5)$$

obtained either from the full-space superoperator propagator $\mathcal{E} = e^{\mathcal{L}T}$ (where $\mathcal{L}$ is the Lindbladian) or, more cheaply, from $d^2 = 64$ master-equation runs with non-Hermitian initial operators $|\psi_i\rangle\langle\psi_j|$ (see `triqq/pedersen.py:choi_on_subspace_via_mesolve`). The process fidelity Equation 3 is then evaluated as the contracted tensor

$$F_{\text{pro}} = \frac{1}{64} \sum_{i,j,k,l} U_0^*[k, i] C[i, j, k, l] U_0[l, j]. \quad (6)$$

The trace-preservation indicator (computational-subspace survival) is the partial trace

$$T_P = \frac{1}{8} \sum_{i,k} C[i, i, k, k]. \quad (7)$$

Phase-corrected variant (recommended for reporting). Many coherent diagonal phases on the 8 computational basis states are removable by the surrounding compiler (single-qubit virtual- Z , or circuit-level tracking of multi-qubit CZ). The **phase-corrected** Pedersen fidelity replaces U_0 in Equation 3 by $D U_0$ and maximises over all diagonal unitaries D :

$$\overline{F}_{\text{PC}} = \max_{D \text{ diag}, |D_{kk}|=1} \overline{F}(\mathcal{E}, D U_0) \quad (8)$$

For a gate that implements U_0 up to a known diagonal correction D (the typical case for Rydberg blockade gates with parasitic V_{cc}), this is the right figure of merit for the *operationally* relevant gate quality.

1.3 Numerical results at $V_{cc}/(2\pi) = -5.54$ MHz

Quantity	CCX gate	OR gate
Computational dim d	8	8
Gate time T	40 ns	320 ns
Lattice spacing a	5.00 μm	5.00 μm
$V_{\text{ct}}/(2\pi)$ (Rb–Cs Förster)	+516.81 MHz	+516.81 MHz

$V_{cc}/(2\pi)$ (Cs–Cs vdW)	−5.54 MHz	−5.54 MHz
Survival T_P	0.996744	0.996173
Basis fidelity $\overline{F}_{\text{basis}}$	0.996620	0.992611
Pedersen raw $\overline{F}$	0.796283	0.180560
Pedersen $\overline{F}_{\text{PC}}$	0.996997	0.992572
Diagonal D structure	single-qubit $Z + \text{CZ}_{c_1 c_2}$	single-qubit $Z + \text{CZ}_{c_1 c_2}$
Infidelity $1 - \overline{F}_{\text{PC}}$	3.00×10^{-3}	7.43×10^{-3}

Per-input basis fidelities $F_k = \langle \pi(k), \mathcal{E}(\rho_{kk}), \pi(k) \rangle$:

Input	Ideal output (CCX / OR)	F_k for CCX	F_k for OR
$ 0, 0, A\rangle$	$ 0, 0, A\rangle / 0, 0, A\rangle$	0.992175	0.998810
$ 0, 0, B\rangle$	$ 0, 0, B\rangle / 0, 0, B\rangle$	0.991581	0.998810
$ 0, 1, A\rangle$	$ 0, 1, A\rangle / 0, 1, B\rangle$	0.996745	0.994251
$ 0, 1, B\rangle$	$ 0, 1, B\rangle / 0, 1, A\rangle$	0.997920	0.994251
$ 1, 0, A\rangle$	$ 1, 0, A\rangle / 1, 0, B\rangle$	0.996745	0.994251
$ 1, 0, B\rangle$	$ 1, 0, B\rangle / 1, 0, A\rangle$	0.997920	0.994251
$ 1, 1, A\rangle$	$ 1, 1, B\rangle / 1, 1, B\rangle$	0.999937	0.983128
$ 1, 1, B\rangle$	$ 1, 1, A\rangle / 1, 1, A\rangle$	0.999937	0.983128
Mean		0.996620	0.992611

Notation reminder: CCX = Toffoli (target flips iff both controls $|1\rangle$); OR = target flips iff at least one control is $|1\rangle$.

1.4 Draft Methods paragraph

To benchmark the gate quality we compute the Haar-averaged average gate fidelity introduced by Pedersen, Mølmer, and Mølmer (**ref: Pedersen 2007**),

$$\overline{F}(\mathcal{E}, U_0) = (dF_{\text{pro}} + 1)/(d + 1), \quad (9)$$

where $\mathcal{E}$ is the simulated CPTP gate and U_0 the ideal target unitary on the $d = 2^n$ -dimensional computational subspace ($n = 3$ for both gates studied here, so $d = 8$). The process fidelity is

$$F_{\text{pro}}(\mathcal{E}, U_0) = 1/d^2 \sum_{i,j=1}^d \langle \pi(i), \mathcal{E}(\rho_{ij}), \pi(j) \rangle, \quad (10)$$

with $\rho_{ij} = |\psi_i\rangle\langle\psi_j|$ the computational operators and $|\pi(i)\rangle = U_0 |\psi_i\rangle$ the ideal output. We evaluate this by computing the rank-4 Choi tensor of $\mathcal{E}$ restricted to the 8-dim computational subspace, via $d^2 = 64$ Lindblad-master-equation runs of the full 36-dim simulator with non-Hermitian initial operators ρ_{ij} (**ref: TriQG codebase**). Decoherence is modelled as spontaneous emission from each Rydberg state with ARC-computed lifetimes $\tau_r = 142.73 \mu\text{s}$ (Cs $|r\rangle$), $\tau_R = 134.87 \mu\text{s}$ (Rb $|R\rangle$), and $\tau_P = 0.131 \mu\text{s}$ (Rb $|P\rangle$) at $T = 300$ K. Coherent diagonal phases on the computational basis are absorbed by virtual-Z frame changes and circuit-level CZ tracking; we therefore report the phase-corrected average gate fidelity $\overline{F}_{\text{PC}} = \max_D \overline{F}(\mathcal{E}, DU_0)$ over all diagonal unitaries D in the computational basis. The bare Pedersen fidelity $\overline{F}$ against U_0 is additionally tabulated in the Supplementary Material.

1.5 Draft Results paragraph

With the parameters of Table 1, we find phase-corrected average gate fidelities of

$$\overline{F}_{\text{PC}}^{\text{CCX}} = 0.997, \quad \overline{F}_{\text{PC}}^{\text{OR}} = 0.993, \quad (11)$$

corresponding to infidelities 3.0×10^{-3} and 7.4×10^{-3} respectively. The dominant contribution in both cases is spontaneous decay during the gate (computational-subspace survival $T_P = 0.997$ for CCX, 0.996 for OR), set by the $0.131 \mu\text{s}$ lifetime of the intermediate Rb $|P\rangle$ state. The diagonal correction absorbed into $\overline{F}_{\text{PC}}$ decomposes into single-qubit virtual- Z rotations and a two-qubit $\text{CZ}_{c_1c_2}$ between the control atoms; the latter is induced by the residual Cs–Cs van der Waals interaction $V_{cc}/(2\pi) = -5.54 \text{ MHz}$ at the operating inter-control distance $r_{\text{AA}} = 5\sqrt{2} \mu\text{m}$. We have verified that no three-body CCZ component contributes at above the 10^{-4} level in either gate.

1.6 Suggested figure: per-input fidelity bar chart

A single panel showing F_k for both gates is the cleanest visual. Use a grouped bar chart with the eight computational inputs along the x-axis (labelled $|0, 0, A\rangle, |0, 0, B\rangle, \dots, |1, 1, B\rangle$) and F_k on the y-axis (range 0.98 to 1.00). One bar per gate in each group, plus a horizontal dashed line at $\overline{F}_{\text{basis}}$ for each gate.

The data values are exactly those tabulated in the per-input table above. The visual story it tells:

- For CCX, F_k is essentially flat at ≈ 0.997 across all inputs except $|0, 0, \cdot\rangle$ (slightly lower at 0.992 from V_{cc} -induced leakage during the joint $|r, r\rangle$ excitation).
- For OR, F_k degrades for the higher-multiplicity branches: 0.999 on $|0, 0, \cdot\rangle$ (controls don't excite to $|r\rangle$), 0.994 on $|0, 1, \cdot\rangle$ and $|1, 0, \cdot\rangle$, 0.983 on $|1, 1, \cdot\rangle$ (both controls in $|r\rangle$ for the longest – maximal $|R\rangle$ decay during the 320 ns Raman pulse).

The figure caption should state the parameters ($V_{cc}/(2\pi) = -5.54 \text{ MHz}$, $a = 5 \mu\text{m}$, $T_{\text{gate}} = 40 \text{ ns} / 320 \text{ ns}$) and the overall $\overline{F}_{\text{PC}}$ for each gate.

1.7 Mandatory disclosures

1. The protocol implements $U_0 \cdot \text{CZ}_{c_1c_2}$ plus local Z rotations, where the CZ comes from the V_{cc} phase on the $|r, r\rangle$ doubly-Rydberg state. Quote the eight optimal phases φ_k^* from the run output explicitly so others can reconstruct D .
2. Decoherence model: spontaneous emission only, with ARC-computed lifetimes at $T = 300 \text{ K}$. No dephasing, no scattering from the trap, no laser noise. State explicitly that the reported number is an **upper bound** on the experimental fidelity for any given hardware.
3. The OR gate uses the EIT-blockade scheme with a smooth super-Gaussian Raman pulse on the target (Farouk et al. 2023); the CCX gate uses a three-square-pulse direct $|A, B\rangle \leftrightarrow |R\rangle$ scheme on Rb. State the protocol explicitly so the reader knows which gate corresponds to which architecture.

Optional supplementary plots:

- $\overline{F}_{\text{PC}}$ vs. lattice spacing a , showing the magic-spacing peaks at $a \approx 3.75, 3.34, 3.12 \mu\text{m}$ where $V_{cc}T_{\text{wait}} = 2\pi N$ for $N = 1, 2, 3$.
- The 8 noiseless branch phases φ_k for both gates as a stem plot, decomposed into the three sources (V_{cc} on $|0, 0, \cdot\rangle$, AC Stark, target X block).

2 The four metrics, equation by equation

Let $\mathcal{E}$ be the gate (a CPTP map on the 36-dim simulator), $\mathcal{S}$ the 8-dim computational subspace, U_0 the ideal 8×8 CCX unitary, $\{|\psi_i\rangle\}_{i=1}^8$ the 8 computational basis kets, and $|\pi(i)\rangle = U_0 |\psi_i\rangle$ the ideal output for input i . Each metric below is a single number computed from the same run.

2.1 Survival T_P – did anything leak out?

Plain-language definition. Run the gate from each of the 8 classical inputs. After the gate, measure how much population remains inside the computational subspace (i.e. is **not** in $|r\rangle$, $|R\rangle$, or $|P\rangle$). Average over the 8 inputs:

$$T_P = \frac{1}{8} \sum_{i=1}^8 \text{Prob}(\text{system in } \mathcal{S} \text{ after gate} \mid \text{started in } |\psi_i\rangle) \quad (12)$$

In formula form, with $\mathcal{E}(\rho)$ the gate’s output density matrix and $\Pi_{\mathcal{S}} = \sum_k |\psi_k\rangle\langle\psi_k|$ the projector onto the computational subspace,

$$T_P = \frac{1}{8} \sum_{i=1}^8 \text{Tr}[\Pi_{\mathcal{S}} \mathcal{E}(|\psi_i\rangle\langle\psi_i|)] \quad (13)$$

What it sees. Population that bled into $|r\rangle$, $|R\rangle$, or $|P\rangle$ during the gate (Rydberg decay).

What it ignores. Whether the population that **did** stay inside $\mathcal{S}$ ended up in the right place, and any phase information.

Your number. $T_P = 0.998129$. So $1 - T_P = 1.87 \times 10^{-3}$ of the population leaked out per gate event.

2.2 Basis fidelity $\overline{F}_{\text{basis}}$ – did each classical input go to the right output?

Plain-language definition. For each of the 8 classical inputs $|\psi_i\rangle$, compute the probability of finishing in the **correct** output state $|\pi(i)\rangle$. Average over the 8 inputs:

$$\overline{F}_{\text{basis}} = \frac{1}{8} \sum_{i=1}^8 \text{Prob}(\text{final state is } |\pi(i)\rangle \mid \text{started in } |\psi_i\rangle) \quad (14)$$

In formula form,

$$\overline{F}_{\text{basis}} = \frac{1}{8} \sum_{i=1}^8 \langle \pi(i) | \mathcal{E}(|\psi_i\rangle\langle\psi_i|) | \pi(i) \rangle \quad (15)$$

This is exactly Yu et al.’s Eq. (7) and the standard “computational-basis-averaged” fidelity reported in most experimental Rydberg papers.

What it sees. Population leakage **and** getting to the wrong computational state (e.g. $|0, 0, A\rangle$ ending up as $|0, 0, B\rangle$).

What it ignores. Phases. If the output is $e^{i\varphi} |\pi(i)\rangle$ for any φ , the basis fidelity is still 1. Coherent diagonal errors – including the spurious CZ_{c₁c₂} from V_{cc} – are completely invisible.

Relation to survival. $\overline{F}_{\text{basis}} \leq T_P$ always, because “landed in the **right** state” is a stricter requirement than “landed in **any** computational state”. The gap $T_P - \overline{F}_{\text{basis}}$ is the probability of the gate shuffling population to the wrong computational state without leaking it out of $\mathcal{S}$.

Your number. $\overline{F}_{\text{basis}} = 0.998011$, so $T_P - \overline{F}_{\text{basis}} = 1.18 \times 10^{-4}$. Almost all of the T_P infidelity is leakage, not logical error.

2.3 Raw Pedersen fidelity $\overline{F}_{\text{raw}}$ – is the gate equal to the literal U_0 ?

Plain-language definition. Average the output-state fidelity over Haar-random pure inputs (not just the 8 classical ones). Coherent superpositions probe phase relationships between branches; if those phases don’t match U_0 , you pay.

The Pedersen / Nielsen-Horodecki formula reduces this to two simple building blocks:

Step 1: process fidelity	$F_{\text{pro}} = \frac{1}{d^2} \text{Tr}(U_0^\dagger U_{\text{eff}}) ^2 \quad (16)$ where $d = 8$ and $U_{\text{eff}} = V^\dagger U_{\text{full}} V$ is the 8x8 projection of the simulator's noiseless propagator onto $\mathcal{S}$ via the isometry V.
Step 2: convert to average fidelity	$\bar{F}_{\text{raw}} = \frac{d F_{\text{pro}} + 1}{d + 1} = \frac{8 F_{\text{pro}} + 1}{9} \quad (17)$ This is Pedersen Eq. (5) – relates F_{pro} to the Haar-averaged pure-state fidelity. For the noisy case, replace $\text{Tr} ^2$ by the Choi-tensor expression in Pedersen Eq. (3); the relation is unchanged.

Geometric picture. When $U_{\text{eff}} \approx D U_0$ for some diagonal $D = \text{diag}(e^{i\varphi_1}, \dots, e^{i\varphi_8})$,

$$F_{\text{pro}} = \frac{1}{64} \left| \sum_{k=1}^8 e^{i\varphi_k} \right|^2. \quad (18)$$

Eight unit vectors on the complex plane. If they all point in the same direction (all φ_k equal), the sum has magnitude 8 and $F_{\text{pro}} = 1$. If they fan out, they cancel destructively and F_{pro} collapses. With Option 0 active and $V_{cc} = 0$, all 8 vectors lie within 0.05π of $+1$, giving $F_{\text{pro}} = 0.996$. With $V_{cc} = -5.54$ MHz, two vectors swing to $\approx +0.4\pi$ and the sum drops, giving $F_{\text{pro}} = 0.77$.

What it sees. Everything – leakage, logical errors, **and** coherent phase patterns. Including phase errors that are removable for free.

What it ignores. Nothing – which is exactly why it is the wrong number to print as a headline.

Your number. $\bar{F}_{\text{raw}} = 0.995625$ at $V_{cc} = 0$ (essentially the same as $\bar{F}_{\text{basis}}$, because there are no phase errors to penalise). At $V_{cc} = -5.54$ MHz, $\bar{F}_{\text{raw}} = 0.796$ – the drop is **entirely** the removable $\text{CZ}_{c_1 c_2}$.

2.4 Phase-corrected Pedersen fidelity $\bar{F}_{\text{PC}}$ – is the gate equal to U_0 up to free corrections?

Plain-language definition. Same as $\bar{F}_{\text{raw}}$, but you are **allowed to redefine the target** as $D U_0$ where D is any 8×8 diagonal unitary. Maximise the fidelity over all such D . The optimum D absorbs all coherent diagonal phase errors of the gate.

In formula form:

The optimisation	$\bar{F}_{\text{PC}} = \max_{D \text{ diagonal}, D_{kk} =1} \text{Pedersen avg fidelity}(\mathcal{E}, D U_0) \quad (19)$
Why it is allowed	A diagonal D in the computational basis is a product of single-qubit Z rotations and multi-qubit CZ/CCZ gates. Single-qubit Z 's are free (virtual- Z frame change). Multi-qubit CZ 's are absorbed by the surrounding Clifford compiler. So if the gate is $D U_0$, the useful part of it is exactly U_0 – the D is bookkeeping.

Decomposition of the diagonal D . Eight phases factor uniquely into:

$$D = e^{i\alpha} \underbrace{(Z^{a_1} \otimes Z^{a_2} \otimes Z^{a_3})}_{\text{free virtual-Z}} \underbrace{\text{CZ}_{c_1 c_2}^{b_1} \text{CZ}_{c_1 t}^{b_2} \text{CZ}_{c_2 t}^{b_3}}_{\text{two-body Z's: 3 params, 1 entangling gate each}} \underbrace{\text{CCZ}_{c_1 c_2 t}^{c_1}}_{\text{three-body Z: 1 param, expensive}} \quad (20)$$

What it sees. Real population errors – leakage and decoherence. Off-diagonal coherent errors that **cannot** be absorbed into a diagonal D .

What it ignores. Diagonal coherent errors (any combination of Z 's, CZ's, CCZ). These are exactly the free-to-correct components.

Your number. $\bar{F}_{\text{PC}} = 0.998230$ at $V_{cc} = 0$, 0.996997 at $V_{cc} = -5.54$ MHz. The latter matches Farouk et al.'s 99.7% almost exactly, on the same footing.

2.5 Visual summary of the four metrics

Error type	T_P	$\bar{F}_{\text{basis}}$	$\bar{F}_{\text{raw}}$	$\bar{F}_{\text{PC}}$
Leakage to $ r\rangle, R\rangle, P\rangle$	×	×	×	×
Wrong state inside $\mathcal{S}$ (e.g. $A \rightarrow B$ flip wrong)	✓	×	×	×
Coherent phase, single-qubit Z on any qubit	✓	✓	×	✓
Coherent phase, two-body CZ	✓	✓	×	✓
Coherent phase, three-body CCZ	✓	✓	×	✓
Off-diagonal coherent error (e.g. wrong axis)	×	×	×	×

Legend: × = punishes this error (number drops); ✓ = blind to this error (number unchanged). The CCZ row in particular shows the difference between $\bar{F}_{\text{raw}}$ (sees everything) and $\bar{F}_{\text{PC}}$ (absorbs the diagonal part).

2.6 The cascade in plain English

Four numbers, four levels of the **same** error budget:

- $1 - T_P$ = **leaked out of the computational subspace** (1.87×10^{-3} in your run).
- $T_P - \bar{F}_{\text{basis}}$ = **stayed inside but landed in the wrong computational state** (1.18×10^{-4}).
- $\bar{F}_{\text{PC}} - \bar{F}_{\text{raw}}$ = **coherent phase error that virtual-Z absorbs for free** (2.6×10^{-3} at $V_{cc} = 0$, 2.0×10^{-1} at $V_{cc} = -5.54$ MHz).
- $1 - \bar{F}_{\text{PC}}$ = **unfixable error: leakage, decoherence, or off-diagonal coherent error** (1.77×10^{-3}).

The last bullet is the only thing that **cannot** be repaired by circuit-level corrections. It is the right “infidelity” to put in a paper.

3 The setup, in one screen

The CCX Hamiltonian (`triqq/hamiltonian.py:132`) is

$$\begin{aligned}
H(t) = & V_{\text{ct}}(|r\rangle\langle r|_{c_1} + |r\rangle\langle r|_{c_2}) \otimes |R\rangle\langle R|_t \\
& + V_{cc} |rr\rangle\langle rr|_{c_1 c_2} \\
& + \Omega_{cc}(t)(|0\rangle\langle r| + |r\rangle\langle 0|)_{c_1, c_2} \\
& + \Omega_{t,1}(t)(|B\rangle\langle R| + |R\rangle\langle B|)_t \\
& + \Omega_{t,2}(t)(|A\rangle\langle R| + |R\rangle\langle A|)_t
\end{aligned} \tag{21}$$

Parameters at $a = 5 \mu\text{m}$ (current run):

Quantity	Value	Origin
$V_{\text{ct}}/(2\pi)$	+516.81 MHz	Rb–Cs Förster, C_3
$V_{cc}/(2\pi)$	0 (set in script)	physical value -5.54 MHz, multiplied by 0 on line 69
$\Omega_{cc}/(2\pi)$	100 MHz	control $ 0\rangle \leftrightarrow r\rangle$ Rabi
$\Omega_t/(2\pi)$	50 MHz	target $ A, B\rangle \leftrightarrow R\rangle$ Rabi

T_{cc}	5 ns	π -pulse on controls
T_t	10 ns	target sub-pulse
T_{gate}	$2T_{cc} + 3T_t = 40$ ns	
τ_r, τ_R, τ_P	143, 135, 0.131 μ s	ARC at 300 K

Pulse-sign convention (current). The control pulse Ω_{cc} is sign-alternating: $+\Omega_{cc}/2$ on the first T_{cc} , $-\Omega_{cc}/2$ on the second. The target sub-pulses now have the pattern $(+, -, +)$: sub-pulses 1 and 3 (driven by `omega_t1`) return $+\frac{\text{amp}}{2}$, while sub-pulse 2 (driven by `omega_t2`) returns $-\frac{\text{amp}}{2}$. This is the result of Option 0 (Section 7.1).

4 Branch-phase budget: three sources, eight numbers

For each computational input $|c_1, c_2, t\rangle$ the noiseless gate produces $e^{i\varphi_k} U_0 |c_1, c_2, t\rangle$ up to small leakage. The phase φ_k is the integrated energy along the trajectory, with three contributions.

4.1 Source 1: V_{cc} on the $|r, r\rangle$ block

Only the $|0, 0, \cdot\rangle$ inputs pump **both** controls to $|r\rangle$ (because Ω_{cc} drives $|0\rangle \leftrightarrow |r\rangle$ on each control, see test `tests/test_hamiltonian.py:285-289`). Time spent in $|r, r\rangle$:

$$T_{rr} = \underbrace{3T_t}_{\text{wait, controls fully in } |rr\rangle} + \underbrace{2 \cdot \frac{3}{8} \cdot T_{cc}}_{\text{average over the two } \pi\text{-pulses}} = 30 + 3.75 = 33.75 \text{ ns} \quad (22)$$

The phase rate on $|r, r\rangle$ is $-V_{cc}$ (Schrödinger evolution e^{-iHt}). Because $V_{cc} < 0$ (Cs-Cs vdW with $C_6 < 0$):

$$\varphi_{cc} = -V_{cc} \cdot T_{rr} = +2\pi \cdot 5.54 \text{ MHz} \cdot 33.75 \text{ ns} = +0.374\pi \quad (23)$$

This is a $\approx +0.37\pi$ phase on every $|0, 0, \cdot\rangle$ branch, verified numerically (Section 4) at the physical $V_{cc}/(2\pi) = -5.54$ MHz. The phase moves to 0 if V_{cc} is multiplied out by hand or if the lattice is tuned to a magic spacing (Section 7.2).

4.2 Source 2: AC-Stark shift from off-resonant target sub-pulses

When the target is blockaded (controls in $|r\rangle$), the sub-pulses that try to drive $|A, B\rangle \leftrightarrow |R\rangle$ at Rabi Ω_t see a detuned $|R\rangle$ shifted by NV_{ct} , where $N \in \{1, 2\}$ is the number of controls in $|r\rangle$. The Stark shift on the driven ground-state level is

$$\delta_{AC} = \frac{\Omega_t^2}{4NV_{ct}} \implies \delta_{AC}/(2\pi) = \frac{50^2}{4N \cdot 517} \text{ MHz} = \frac{1.21}{N} \text{ MHz} \quad (24)$$

Multiplied by the 10 ns each sub-pulse lasts, this is $+0.024\pi/N$ per actuated sub-pulse on the level being driven. **Sign-blind in Ω_t** , so unaffected by Option 0. Sub-pulses 1 and 3 drive $|B\rangle$, sub-pulse 2 drives $|A\rangle$.

4.3 Source 3: target rotation $+X$ on the $|1, 1, \cdot\rangle$ block (post Option 0)

For $|1, 1, \cdot\rangle$ inputs, the controls stay in $|1\rangle$ and the three target sub-pulses fire freely. With the **current** signs $(s_1, s_2, s_3) = (+, -, +)$, each path through $|R\rangle$ picks up **one** factor of $-i$ and **one** factor of $+i$:

$$\begin{aligned} |1, 1, A\rangle &\xrightarrow{\text{sp } 2, -\Omega} +i|1, 1, R\rangle \xrightarrow{\text{sp } 3, +\Omega} (+i)(-i)|1, 1, B\rangle = +|1, 1, B\rangle \\ |1, 1, B\rangle &\xrightarrow{\text{sp } 1, +\Omega} -i|1, 1, R\rangle \xrightarrow{\text{sp } 2, -\Omega} (-i)(+i)|1, 1, A\rangle = +|1, 1, A\rangle \end{aligned}$$

so $U_{\text{tgt}} = +X$, a clean Toffoli action. The diagonal of $U_{\text{eff}} U_0^\dagger$ on $|1, 1, \cdot\rangle$ is $+1$, contributing a phase 0. **No spurious CZ.** Verified by the rerun: `branch_phases_pi[6] = branch_phases_pi[7] = 0.0000`.

If ω_{t2} were restored to $+\frac{\text{amp}}{2}$, the action becomes $U_{\text{tgt}} = -X$ and the $|1, 1, \cdot\rangle$ branch picks up the notorious $+\pi$ phase that costs 0.7 in $\bar{F}_{\text{raw}}$.

4.4 The complete prediction vs. the rerun

Branch	V_{cc}	Stark	X	Predicted	Simulated $V_{cc} = 0$
$ 0, 0, A\rangle$	0	$+0.012\pi$	0	$+0.012\pi$	$+0.0119\pi$
$ 0, 0, B\rangle$	0	$+0.024\pi$	0	$+0.024\pi$	$+0.0233\pi$
$ 0, 1, A\rangle$	0	$+0.024\pi$	0	$+0.024\pi$	$+0.0234\pi$
$ 0, 1, B\rangle$	0	$+0.048\pi$	0	$+0.048\pi$	$+0.0476\pi$
$ 1, 0, A\rangle$	0	$+0.024\pi$	0	$+0.024\pi$	$+0.0234\pi$
$ 1, 0, B\rangle$	0	$+0.048\pi$	0	$+0.048\pi$	$+0.0476\pi$
$ 1, 1, A\rangle$	0	0	0	0	$+0.0000\pi$
$ 1, 1, B\rangle$	0	0	0	0	$+0.0000\pi$

Every branch is reproduced to better than 0.001π . With V_{cc} restored, the $|0, 0, \cdot\rangle$ branches shift to $+0.387\pi$ and $+0.398\pi$ (also confirmed numerically), exactly matching the original analysis with $\varphi_{cc} = +0.374\pi$.

5 Numerical comparison of the four metrics across runs

Metric	$V_{cc} = 0$ (current)	$V_{cc} = -5.54$ MHz	Original (no Option 0)
T_P	0.998129	0.996744	0.996744
$\bar{F}_{\text{basis}}$	0.998011	0.996626	0.996626
$\bar{F}_{\text{raw}}$	0.995625	0.796283	0.282014
$\bar{F}_{\text{PC}}$	0.998230	0.996997	0.996997
gap PC – raw	0.0026	0.201	0.715

Reading the table column by column:

- Current column ($V_{cc} = 0$, Option 0 applied):** all four numbers within 3×10^{-3} . Means there is essentially no removable diagonal error, **and** very little leakage. This is the best-case theoretical limit for the protocol.
- Physical V_{cc} column:** T_P , $\bar{F}_{\text{basis}}$, $\bar{F}_{\text{PC}}$ all stay near 0.997 (V_{cc} costs only 1.4×10^{-3} of leakage on the basis number). But $\bar{F}_{\text{raw}}$ drops to 0.80. The full 0.20 gap is the removable $\text{CZ}_{c_1 c_2}$ from V_{cc} – pure bookkeeping.
- Original column (no Option 0):** $\bar{F}_{\text{basis}}$ and $\bar{F}_{\text{PC}}$ identical to the physical- V_{cc} column above (Option 0 changes only a phase that PC absorbs). $\bar{F}_{\text{raw}}$ collapses to 0.28 because the $+\pi$ on $|1, 1, \cdot\rangle$ adds a **second** removable $\text{CZ}_{c_1 c_2}$ on top of the V_{cc} one.

The story across the row $\bar{F}_{\text{raw}}$: $0.282 \rightarrow 0.796 \rightarrow 0.996$ as you (i) flip $\omega_{t,2}$ sign, then (ii) zero out V_{cc} . Both changes are removable – the underlying gate quality (measured by $\bar{F}_{\text{PC}}$) only moves from 0.997 to 0.998.

5.1 Which part of the diagonal D is removable for free?

The optimal D (the diagonal post-rotation that maximises the phase-corrected fidelity) decomposes uniquely into single-qubit Z , two-qubit CZ , and three-qubit CCZ components. Single-qubit Z 's are free on every platform (virtual- Z); two- and three-qubit phases require additional gates. For the current run, the eight optimal phases (in units of π) are:

$$\varphi^* = [+0.717, -1.272, +0.729, +0.753, +0.729, -1.247, +0.705, -1.295] \quad (25)$$

With $V_{cc} = 0$, this D is dominated by a single-qubit Z on the target and a global phase – both free. With $V_{cc} = -5.54$ MHz restored, the additional structure is a $CZ_{c_1c_2}$; that is **not** free locally, but it commutes with the surrounding Clifford frame and is absorbed by circuit-level tracking (zero gate cost in fault-tolerant compilation). **The CCZ component remains negligible** at both V_{cc} settings – this is a clean Toffoli, not a noisy three-body unitary.

6 How Farouk et al. handled the same physics

The paper’s own diagnosis is unambiguous (p. 14, after Fig. 9):

“In Figure 9b, we study a non-realistic case where we neglected the interaction between control atoms... the maximum value of fidelity becomes possible for a wider range of inter-atomic distances. This case proves that the destructive pattern in system dynamics is a direct result of V_{CC} .”

Their Fig. 9a (realistic, $V_{CC} \neq 0$) shows a 99.7% peak only in a **narrow strip** near $R_{CT} \approx 6 \mu\text{m}$ with $\Omega_c/\Omega_p > 2.5$. Fig. 9b (artificial, $V_{CC} = 0$) shows a much wider plateau. They scanned the $(R_{CT}, \Omega_c/\Omega_p)$ plane and reported the best point. **Setting $V_{cc} = 0$ in our simulation is the equivalent of standing on Farouk’s idealised Fig. 9b plateau.**

6.1 Their sweet spot is a magic-spacing condition

With Cs $C_6 = 2\pi \cdot 2364 \text{ GHz } \mu\text{m}^6$ and their geometry $R_{CC} = (2/\sqrt{5})R_{CT} \approx 0.894R_{CT}$:

$$V_{CC}/(2\pi) \text{ at } R_{CC} \approx 5.36 \mu\text{m} \rightarrow 100 \text{ MHz} \quad (26)$$

(matches Fig. 9c). With gate duration $\tau \approx 1.5 \mu\text{s}$:

$$\frac{V_{CC} \cdot \tau}{2\pi} \approx 100 \text{ MHz} \cdot 1.5 \mu\text{s} = 150 \rightarrow \text{integer} \quad (27)$$

The $|1, 1\rangle$ branch makes **150 full revolutions** under V_{CC} during the gate and lands back on its starting phase. Move R_{CT} to $12 \mu\text{m}$ and V_{CC} drops to ≈ 1.6 MHz; then $V_{CC}\tau/(2\pi) \approx 2.36$, off by $0.36 \cdot 2\pi \approx 130^\circ$, and fidelity collapses — exactly the “sharp drop” they describe at $8 \mu\text{m} < R_{CT} < 12 \mu\text{m}$.

6.2 Apples-to-apples comparison

Farouk’s reported number is **state fidelity for a single superposition input**:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle |AA\rangle + |11\rangle |AA\rangle) \rightarrow \frac{1}{\sqrt{2}}(|00\rangle |AA\rangle + |11\rangle |BB\rangle) \quad (28)$$

$$F = |\langle \psi^{\text{target}} | \psi^{\text{actual}} \rangle|^2 = \frac{1 + \cos \varphi_{cc}}{2} \quad (29)$$

This is exactly the kind of quantity that is **sensitive** to the V_{cc} phase. Farouk’s 99.7% requires $\cos \varphi_{cc} \approx 1$ modulo 2π – the magic-spacing condition.

The TriQG CCX with Option 0 and V_{cc} artificially zeroed reaches $\bar{F}_{PC} = 0.998$, slightly **better** than Farouk because the gate is 40 ns instead of $1.5 \mu\text{s}$ and pays less Rb $|R\rangle$ decay. With Option 0 only and the physical V_{cc} , the apples-to-apples score is $\bar{F}_{PC} = 0.997$ – equal to Farouk’s. The “raw” 0.80 in that case is the same diagonal $CZ_{c_1c_2}$ that Farouk hides inside the integer $V_{CC}\tau$.

7 Four engineering options to remove the residual phase mismatch

These are presented in increasing order of **physics cost**. Option 0 is already in place in the current code; Options 1–4 attack the residual V_{cc} contribution.

7.1 Option 0 (free, applied): Flip the sign of $\omega_{t,1}$

Status: applied. `triqq/pulses.py:omega_t2` returns $-\frac{\text{amp}}{2}$. The $|1, 1, \cdot\rangle$ branch phase moved from π to 0.

Why this works. The branch sign acquired by $|1, 1, A\rangle$ through U_{tgt} is the product of the two sub-pulses that **actually couple to its trajectory** (sub-pulses 2 and 3). The branch sign for $|1, 1, B\rangle$ is the product of sub-pulses 1 and 2. With sign vector (s_1, s_2, s_3) ,

$$U_{\text{tgt}} = \begin{pmatrix} 0 & -s_1 s_2 \\ -s_2 s_3 & 0 \end{pmatrix}. \quad (30)$$

A clean $+X$ requires $s_1 s_2 = s_2 s_3 = -1$, which is satisfied by $(s_1, s_2, s_3) = (+, -, +)$ (current) or $(-, +, -)$. Independently flipping **one** outer sub-pulse gives an asymmetric $\pm i\sigma_y$ and does **not** help, because half of $\omega_{t,1}$ is useless on any given input trajectory: sub-pulse 1 is dead for $|\cdot, \cdot, A\rangle$ inputs and sub-pulse 3 is dead for $|\cdot, \cdot, B\rangle$ inputs.

What it bought. Eliminated the $+\pi$ on the $|1, 1, \cdot\rangle$ block, i.e. removed the spurious $\text{CZ}_{c_1 c_2}$. Confirmed improvements (with $V_{cc} = 0$ idealisation):

$$\bar{F}_{\text{raw}} : 0.282 \rightarrow 0.996, \quad \bar{F}_{\text{basis}} : 0.997 \rightarrow 0.998, \quad \bar{F}_{\text{PC}} : 0.997 \rightarrow 0.998 \quad (31)$$

What it does not buy. The V_{cc} -induced $\approx +0.37\pi$ on $|0, 0, \cdot\rangle$ (visible at $V_{cc} = -5.54$ MHz, where $\bar{F}_{\text{raw}}$ is still only 0.80) and the AC-Stark $\approx 0.05\pi$ on the $|0, 1\rangle, |1, 0\rangle$ branches. Those need Options 1–4.

Cost: zero. No additional pulses, no extra time, no extra decoherence.

7.2 Option 1: Magic-spacing trick (Farouk’s solution, ported)

What: Choose the lattice spacing a so that $|V_{cc}| \cdot T_{\text{wait}} = 2\pi N$ for some small integer N .

The condition. With current $T_{\text{wait}} \approx 30$ ns and $C_6 = -693$ GHz μm^6 , requiring

$$|V_{cc}| \frac{1}{2\pi} = \frac{N}{T_{\text{wait}}} = N \cdot 33.3 \text{ MHz} \rightarrow R_{\text{AA}} = \left(\frac{|C_6| \cdot T_{\text{wait}}}{2\pi N} \right)^{1/6} \quad (32)$$

gives the table:

N	$ V_{cc} /(2\pi)$	R_{AA}	Lattice $a = R_{\text{AA}}/\sqrt{2}$
1	33.3 MHz	5.31 μm	3.75 μm
2	66.7 MHz	4.73 μm	3.34 μm
3	100.0 MHz	4.41 μm	3.12 μm

Bonus. Tightening the lattice also shrinks $R_{\text{DA}} = a/\sqrt{2}$ and increases $V_{\text{ct}} = C_3/R_{\text{DA}}^3$. At $a = 3.75$ μm , $V_{\text{ct}}/(2\pi)$ jumps from 517 MHz to ≈ 1.23 GHz, which **reduces** finite-blockade leakage as $(V_{\text{ct}}/\Omega_t)^{-2}$ – roughly a $5 \times$ reduction in the dominant residual error.

Tolerance. The phase deviation from the magic point is $\Delta\varphi_{cc} = (\partial V_{cc}/\partial R_{\text{AA}}) \cdot \Delta R_{\text{AA}} \cdot T_{\text{wait}} = -6V_{cc}\Delta R_{\text{AA}}/R_{\text{AA}} \cdot T_{\text{wait}}$. For $|\Delta\varphi_{cc}| < \pi/100$ ($> 99.95\%$ overlap), one needs $|\Delta R_{\text{AA}}|/R_{\text{AA}} < 10^{-2}/(6N)$. At $N = 1$, $a = 3.75$ μm , this is ≈ 17 nm tolerance on the inter-atomic distance — within reach of optical-tweezer spacing calibration but not free.

Caveats.

- **Ground-state interaction.** Below $\approx 3 \mu\text{m}$, the long-range ground-state forces (mostly van der Waals between $|6S_{1/2}\rangle$ Cs atoms) start lifting the lattice approximation; you will need to verify that the trap depth and loading temperature still allow sub-Lamb-Dicke confinement.
- **Cross-talk.** Tighter a shrinks R_{DA} , which can put nearby Rb–Cs pairs into a regime where leakage to off-resonant Förster channels becomes significant. The Ireland et al. 2024 channel list must be re-checked at the new geometry.
- **Magic spacing is single-frequency.** It only kills $V_{cc}\tau$, not other coherent shifts (e.g. AC Stark on the target). Those remain at the few-percent level.
- **Population leakage from V_{cc} remains.** The current rerun shows that even the **basis** fidelity drops by 1.4×10^{-3} when $V_{cc} = -5.54 \text{ MHz}$ is restored. This is **not** a phase, but a real population leak from V_{cc} detuning the $|0,0\rangle \rightarrow |r,r\rangle$ excitation during the control π -pulses (see Section 8 below). Magic spacing kills the $|r,r\rangle$ phase but does **not** cure this leakage; the only way to suppress it fully is $|V_{cc}|/\Omega_{cc} \ll 1$, which automatically holds at the small- V_{cc} end of the magic-spacing table ($N = 1, a = 3.75 \mu\text{m}$: $|V_{cc}|/\Omega_{cc} = \frac{33.3}{100} = 0.33$, marginal) and fails at high N .

What it buys. $\bar{F}_{\text{raw}} \rightarrow \bar{F}_{\text{PC}}$ on the $|0,0,\cdot\rangle$ branch, eliminating the single largest source of phase mismatch. Combined with Option 0, $\bar{F}_{\text{raw}} \approx \bar{F}_{\text{PC}} \approx 0.997$ at the physical V_{cc} , or ≈ 0.998 with the additional 1.4×10^{-3} leakage suppression at small $|V_{cc}|/\Omega_{cc}$.

7.3 Option 2: Stretch T_{wait} at fixed lattice

What: Hold $a = 5 \mu\text{m}$ (so $V_{cc}/(2\pi) = -5.54 \text{ MHz}$). Pick T_{wait} so that $|V_{cc}| \cdot T_{\text{wait}} = 2\pi N$:

$$T_{\text{wait}}^{\text{magic}} = N / |V_{cc}|/(2\pi) = N \cdot 180.5 \text{ ns} \quad (33)$$

The smallest magic wait is $\approx 180 \text{ ns}$, $6 \times$ longer than the current 30 ns . Total gate time grows from 40 ns to $2T_{cc} + 180 = 190 \text{ ns}$.

What you pay. Extra time spent in the lossy Rydberg states. The incremental decoherence cost is

$$1 - F_{\text{survival}} \approx (\Delta T_{\text{wait}})(\gamma_R + \gamma_P P_R) \quad (34)$$

With $\Delta T_{\text{wait}} = 150 \text{ ns}$, $\tau_R = 135 \mu\text{s}$ and assuming the target spends $\approx O(1)$ of its time in $|R\rangle$ during the extra wait (mostly **not**, since the target sub-pulses are not extended), the dominant cost is $|R\rangle$ decay during the intermediate sub-pulses, $\approx 150/135000 = 1.1 \times 10^{-3}$ extra infidelity. **This is a real loss**, comparable to the basis infidelity itself.

Variant. Instead of stretching the wait, **hold the target sub-pulses at full duration but insert idle gaps between them** until $T_{\text{wait}} = 180 \text{ ns}$. The target sub-pulse decoherence is unchanged; only the controls' $|r\rangle$ population decays during the extra wait. With $\tau_r = 143 \mu\text{s}$, the cost drops to $\approx 2 \cdot 150/143000 \approx 2 \times 10^{-3}$ (factor of 2 for two controls in $|r\rangle$). Still costly.

Verdict. Cheaper than Option 1 (no lattice rebuild), but trades a coherent error for a population error of similar magnitude. Useful mostly as a quick numerical experiment to **verify** the magic-spacing mechanism, not as a final design.

7.4 Option 3: Accept V_{cc} as a free CZ and report

$\bar{F}_{\text{PC}}$

What: Stop computing $\bar{F}_{\text{Pedersen,raw}}$ as the figure of merit. Report either $\bar{F}_{\text{basis}}$ or $\bar{F}_{\text{PC}}$, both of which exceed 0.997 even at the physical V_{cc} .

Why this is legitimate. The “missing” infidelity is a known unitary D acting diagonally on the computational basis. Decompose:

$$D = e^{i\alpha} (Z^{a_1} \otimes Z^{a_2} \otimes Z^{a_3})_{\text{local}} (CZ_{c_1 c_2})^{b_1} (CZ_{c_1 t})^{b_2} (CZ_{c_2 t})^{b_3} (CCZ_{c_1 c_2 t})^{c_1} \quad (35)$$

For your CCX, the 8 optimal phases factor (to within numerical precision) as

$$D \approx D_{\text{local } Z} \cdot \left(\text{CZ}_{c_1 c_2} \right)_{\text{from } V_{cc}} \cdot D_{\text{small Stark } Z} \quad (36)$$

with no measurable CCZ component (verified at both $V_{cc} = 0$ and $V_{cc} = -5.54$ MHz). So:

- the local Z 's are absorbed by virtual- Z on subsequent gates (cost = 0);
- the V_{cc} -induced $\text{CZ}_{c_1 c_2}$ is absorbed by **one** software CZ tracked through the surrounding Clifford layer (cost = 0 in fault-tolerant compilation, since CZ's commute through the Clifford frame);
- the small two-body Z 's from Stark are at the 10^{-4} level and can be dropped.

What it buys. Honest fidelity number that matches the gate's operational quality. Apples-to-apples with Farouk: their reported 99.7% is also a “phase-tuned” number.

What it does not buy. If your downstream circuit cannot absorb a $\text{CZ}_{c_1 c_2}$ for free (e.g. you're benchmarking the CCX as a standalone process tomography target with no surrounding Clifford layer), this option is unavailable and you need Option 0 + 1 instead. Also: the 1.4×10^{-3} population leakage from V_{cc} detuning the $|0, 0\rangle \rightarrow |r, r\rangle$ excitation is **not** a Z phase and is **not** absorbable here. To regain that, you still need Option 1.

7.5 Option 4: Hahn echo on V_{cc} within the gate

What: Insert a midway control operation that swaps $|r, r\rangle \leftrightarrow |0, 0\rangle$ at $T_{\text{wait}}/2$, so that V_{cc} accumulates $+\varphi$ on the first half and $-\varphi$ on the second.

The pulse sequence. Modify the protocol to

$$+\pi_{cc} \rightarrow T_{\text{wait}}/2 \rightarrow 2\pi_{cc} \rightarrow T_{\text{wait}}/2 \rightarrow -\pi_{cc} \quad (37)$$

The middle 2π -pulse on each control returns it to $|0\rangle$ and re-excites it to $|r\rangle$, swapping the population trajectory through a momentary $|0, 0\rangle$ window. Total $V_{cc}\tau$ contributions cancel.

What you pay. Two extra control π -pulses (or one 2π -pulse). Time: $+2T_{cc} = 10$ ns. Decoherence: an extra $\approx 10/143000 \approx 7 \times 10^{-5}$ from $|r\rangle$ decay – **negligible**.

Subtleties.

- The mid-sequence 2π -pulse must occur **during the target wait window**, but the target is mid-Raman / mid-Rabi at that point. The swap will introduce a transient detuning on the target sub-pulse 2 unless the controls are fully de-excited and re-excited cleanly.
- For sub-pulse 2 (which drives $|A\rangle \leftrightarrow |R\rangle$) the target is in $|R\rangle$ during $\approx 50\%$ of the sub-pulse; during that time, swapping the controls in/out of $|r\rangle$ modulates the blockade detuning V_{ct} , which is not cancelled by the echo. Numerical check required.
- The clean version is to insert the echo **between** sub-pulses, e.g. after sub-pulse 1 and before sub-pulse 3. That breaks the symmetry of the target pulse-area and may need re-optimisation of $\Omega_t T_t = \pi$.

What it buys. Removes the V_{cc} phase **exactly**, independent of lattice spacing. Combined with Option 0, gives $\bar{F}_{\text{raw}} \approx \bar{F}_{\text{PC}} \approx 0.997$ at the **current** lattice $a = 5$ μm .

What it does not buy. Same as Option 1: the Stark contribution is unchanged, and the 1.4×10^{-3} V_{cc} -induced population leakage during the **control** π -pulses is also unchanged (the echo is inside the target wait window, not inside the control pulses).

8 New finding from the rerun: V_{cc} -induced leakage during the control pulses

The original analysis claimed V_{cc} was a pure Z phase on the $|r, r\rangle$ block, hence invisible to $\bar{F}_{\text{basis}}$. The rerun shows this is **almost** true but not quite: at the physical $V_{cc} = -5.54$ MHz, $\bar{F}_{\text{basis}}$ drops by 1.4×10^{-3} relative to the $V_{cc} = 0$ baseline. Source:

The two-control system under the Hamiltonian $H_{cc} + V_{cc}P_{rr}$, where H_{cc} is the symmetric Rabi drive on both controls and P_{rr} is the projector onto the doubly-Rydberg state, does **not** factorise into two independent single-atom rotations whenever V_{cc} is nonzero. In the symmetric two-atom subspace, the Hamiltonian becomes a 3-level cascade with effective Rabi $\sqrt{2}\Omega_{cc}$ between the ground state, the symmetric singly-excited state, and the doubly-excited state, with the doubly-excited level detuned by V_{cc} . A nominally π -pulse on each individual control no longer fully transfers the population from ground to doubly-excited; the transfer infidelity is

$$\varepsilon_{\text{transfer}} \approx \frac{V_{cc}^2}{\Omega_{cc}^2} \approx 3 \times 10^{-3} \quad (38)$$

per control pair excitation event, halved over the two excitations (in and out) because the protocol is a near-echo. The observed drop of 1.4×10^{-3} matches this estimate to within a factor of 2.

Implication for Option 1. Magic spacing cures the V_{cc} phase, but if the magnitude of V_{cc} is large, the population leakage during the control pulses can dominate. The optimum is small V_{cc} (magic spacing at the smallest N) and large Ω_{cc} such that $(V_{cc} / \Omega_{cc})^2 < 1e-3$, e.g. $\Omega_{cc} > 5 |V_{cc}|$. The idealised limit with $V_{cc} = 0$ is the strict upper bound on what any magic-spacing realisation can achieve.

9 Recommendation

Option	Cost	$\overline{F}_{\text{raw}}$	$\overline{F}_{\text{PC}}$	Recommended for
Pre-Option-0 baseline	–	0.282	0.997	diagnostics only
Option 0 only	zero	0.796	0.997	<i>done already</i>
Option 0 + $V_{cc}=0$ (idealised)	n/a	0.996	0.998	theoretical bound
0 + 1: magic spacing	hardware	≈ 0.996	≈ 0.998	final design
0 + 2: stretch T_{wait}	10^{-3} extra	≈ 0.99	≈ 0.996	verification only
0 + 3: report $\overline{F}_{\text{PC}}$	bookkeeping	0.796	0.997	comparing to Farouk
0 + 4: V_{cc} echo	10^{-5} extra	≈ 0.997	≈ 0.997	if lattice is fixed

Bottom line. Option 0 is in. The current $\overline{F}_{\text{PC}} = 0.998$ with $V_{cc} = 0$ is the operational ceiling of this protocol at $a = 5 \mu\text{m}$ and $T_{\text{gate}} = 40 \text{ ns}$. To approach it with the **physical** V_{cc} in place, choose:

1. **For publication numbers:** Option 3 – report $\overline{F}_{\text{PC}} = 0.997$ honestly and document the $\text{CZ}_{c_1c_2}$ absorption. Apples-to-apples with Farouk.
2. **For raw-number performance:** Option 1 – shrink the lattice to $a = 3.75 \mu\text{m}$ to land on $N = 1$ magic spacing. This also doubles V_{et} and lowers blockade leakage. Expected $\overline{F}_{\text{raw}} \approx 0.997$.
3. **If the lattice is locked at $a = 5 \mu\text{m}$:** Option 4 – insert a V_{cc} Hahn echo. Cost is one extra 2π -pulse on the controls, $\approx 10^{-5}$ extra decoherence.

10 Verification protocol — STATUS

Test	Predicted	Status
Option 0 verification (ω_{t2} flipped)	$ 1, 1, \cdot\rangle$ branch phase: $\pi \rightarrow 0$. $\overline{F}_{\text{raw}}: 0.282 \rightarrow 0.80$	confirmed
Source 1 isolation ($V_{cc} = 0$)	$ 0, 0, \cdot\rangle$ branch phase: $0.39\pi \rightarrow 0.012\pi$	confirmed

Source 2 (Stark) prediction	$ \cdot, \cdot, A\rangle$ phase: $0.024\pi/N$ per A-driving sub-pulse	confirmed within 0.001π
Combined (Option 0 + V_cc=0)	$\bar{F}_{\text{raw}} \approx 0.996$	confirmed: 0.9956
V_cc population leakage	$\Delta(1 - \bar{F}_{\text{basis}}) \approx \frac{(\frac{V_{cc}}{\Omega_{cc}})^2}{2} \approx 1.5 \times 10^{-3}$	confirmed: 1.4×10^{-3}
Option 1 verification ($a = 3.75 \mu\text{m}$)	$\bar{F}_{\text{raw}} \rightarrow \bar{F}_{\text{PC}}$ at the physical V_{cc}	pending – requires lattice-distance scan

All five completed predictions in the original analysis are confirmed to within stated tolerances. The remaining lattice scan (Option 1) is the next experimental step.

Source script: `examples/Average_fidelity_Vcc_newenergy/ccx_pedersen_average_fidelity.py` (line 69: `V_cc_MHz = 1000.0 * C6-CsCs / r_AA**6 * 0`).

Pulse definitions: `trigg/pulses.py` (`omega_t2` returns `-amp / 2`).

Branch-phase analysis: `examples/Average_fidelity_Vcc_newenergy/branch_phase_analysis.py` (uses physical V_{cc} , not zeroed).

Reference: M. Farouk et al., *Photonics* **10**, 1280 (2023), Sec. 5 and Fig. 9.

Pedersen formalism: L. H. Pedersen, N. M. Møller, K. Mølmer, *Phys. Lett. A* **367**, 47 (2007).

Rerun timestamp: 2026-04-27.